

```
In [1]: import sqlite3
import pandas as pd

con = sqlite3.connect("module6_lab2.db")
cur = con.cursor()

print("Database connected")
```

Database connected

```
In [2]: cur.execute("""
CREATE TABLE IF NOT EXISTS customers (
    id INTEGER PRIMARY KEY,
    name TEXT,
    city TEXT,
    total_spent REAL
)
""")

con.commit()

print("Customers table created")
```

Customers table created

MySQL to SQLite Note

SQLite uses a simpler data type system than MySQL. I used INTEGER, TEXT, and REAL data types when creating the customers table.

```
In [3]: customers = [
    (1, "Olivia", "San Diego", 125.50),
    (2, "Ethan", "Sacramento", 210.75),
    (3, "Chloe", "San Francisco", 89.99),
    (4, "Noah", "Fresno", 340.20),
    (5, "Grace", "San Jose", 175.00),
    (6, "Lucas", "Riverside", 95.40),
    (7, "Mia", "Anaheim", 260.30),
    (8, "Henry", "Santa Barbara", 145.80)
]

cur.executemany("""
INSERT OR REPLACE INTO customers (id, name, city, total_spent)
VALUES (?, ?, ?, ?)
""", customers)

con.commit()

print("8 customers inserted")
```

8 customers inserted

```
In [4]: df = pd.read_sql_query("SELECT * FROM customers", con)
```

df

```
Out [4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <thead> <th></th> <th>id</th> <th>name</th> <th>city</th>
<th>total_spent</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody>
</tbody>
```

1	Olivia	San Diego	125.50
2	Ethan	Sacramento	210.75
3	Chloe	San Francisco	89.99
4	Noah	Fresno	340.20
5	Grace	San Jose	175.00
6	Lucas	Riverside	95.40
7	Mia	Anaheim	260.30
8	Henry	Santa Barbara	145.80

MySQL to SQLite Note

SQLite uses a simpler data type system than MySQL. I used INTEGER, TEXT, and REAL data types when creating the customers table.

```
In [5]: # Query 1: WHERE
query1 = """
SELECT *
FROM customers
WHERE total_spent > 150;
"""

result1 = pd.read_sql_query(query1, con)
result1
```

```
Out [5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <thead> <th></th> <th>id</th> <th>name</th> <th>city</th>
<th>total_spent</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> </tbody> <tbody></tbody>
```

2	Ethan	Sacramento	210.75
4	Noah	Fresno	340.20
5	Grace	San Jose	175.00
7	Mia	Anaheim	260.30

```
In [6]: # Query 2: ORDER BY
query2 = """
SELECT *
FROM customers
ORDER BY total_spent DESC;
"""

result2 = pd.read_sql_query(query2, con)
result2
```

```
Out [6]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;
} </style> <thead> <th></th> <th>id</th> <th>name</th> <th>city</th>
<th>total_spent</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody>
</tbody>
```

4	Noah	Fresno	340.20
7	Mia	Anaheim	260.30
2	Ethan	Sacramento	210.75
5	Grace	San Jose	175.00
8	Henry	Santa Barbara	145.80
1	Olivia	San Diego	125.50
6	Lucas	Riverside	95.40
3	Chloe	San Francisco	89.99

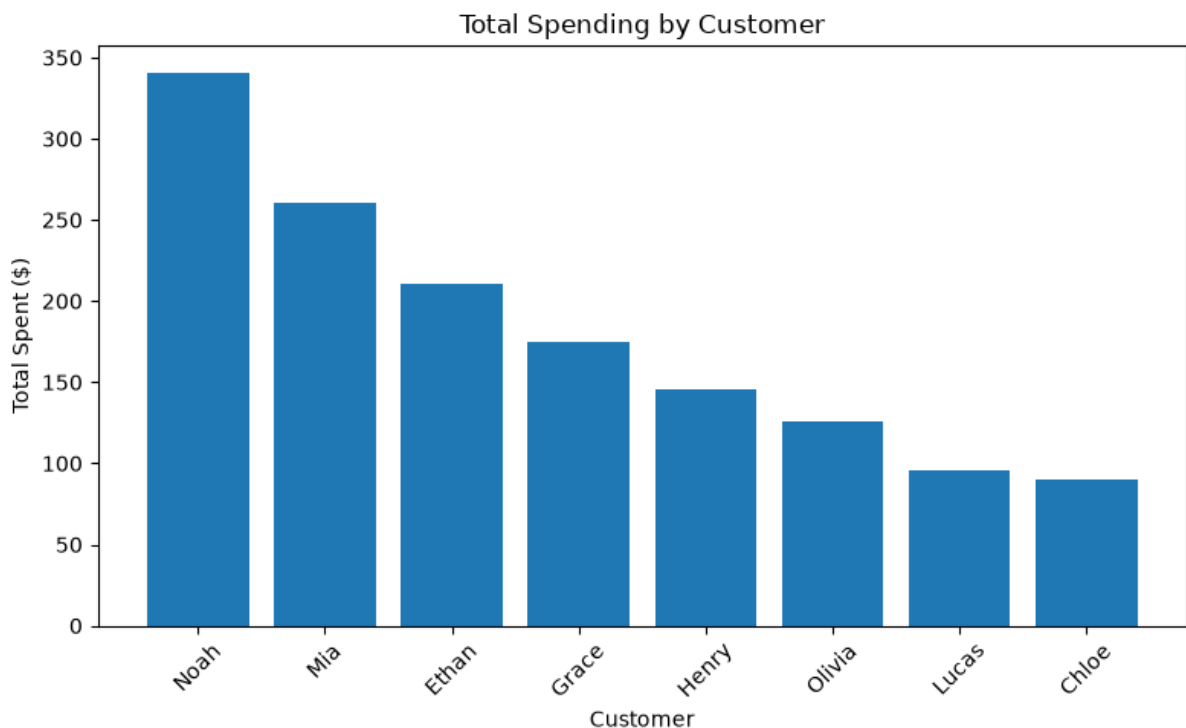
```
In [7]: # Query 3: LIKE
query3 = """
SELECT *
FROM customers
WHERE name LIKE '%a%';
"""

result3 = pd.read_sql_query(query3, con)
result3
```

```
Out [7]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }  
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }  
</style> <thead> <th></th> <th>id</th> <th>name</th> <th>city</th>  
<th>total_spent</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>  
<th>3</th> <th>4</th> <th>5</th> </tbody> <tbody></tbody>
```

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6	Lucas	Riverside	95.40
7	Mia	Anaheim	260.30

```
In [8]: import matplotlib.pyplot as plt  
  
# Chart from Query 2 result  
plt.figure(figsize=(8, 5))  
plt.bar(result2["name"], result2["total_spent"])  
  
plt.title("Total Spending by Customer")  
plt.xlabel("Customer")  
plt.ylabel("Total Spent ($)")  
plt.xticks(rotation=45)  
plt.tight_layout()  
  
plt.show()
```



In []: